



OCUS: A game-theoretic approach to optimal UAV coalitions in UAV-as-a-Service platforms

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ABSTRACT

This paper presents OCUS, an optimal coalition formation scheme designed to enhance the performance of UAV-as-a-Service (UaaS) platforms for Internet of Things (IoT) applications. UaaS platforms typically involve a set of heterogeneous UAVs owned by different entities, serving IoT applications such as agriculture, traffic monitoring, and surveillance often requires multiple UAVs with diverse sensing, processing, and communication capabilities. Thus, forming an optimal coalition of these UAVs is essential for efficient resource utilization and application performance. Despite the growing interest in UAV-based services, existing literature lacks coalition formation schemes tailored for UaaS platform. To address this gap, OCUS leverages a coalition game-theoretic approach, where each UAV acts as a player in the game. The scheme introduces a selection score, which evaluates UAV owners based on reputation, cost competitiveness, and task allocation fairness. Using a payoff-driven method, OCUS promotes coalition stability and maximizes utility for UaaS platforms. The payoff function considers utility, reputation, task allocation fairness, costs, and equilibrium constraints. By employing Nash Equilibrium (NE) principles, OCUS ensures stable coalitions, discouraging UAVs from switching coalitions for higher payoffs. OCUS optimizes coalition formation by maximizing task completion rates, minimizing energy consumption, and ensuring fair and stable UAV participation through game-theoretic modeling. To evaluate OCUS, simulation experiments were conducted over 100 iterations involving 1 to 25 heterogeneous UAVs across a 50×50 unit² area, with varying distances and task counts (1 – 15). Simulation results demonstrate that OCUS achieves 25%–30% more stable coalitions compared to state-of-the-art approaches. Furthermore, NE principles enable OCUS to form coalitions that balance efficiency and fairness, advancing the capabilities of UaaS platforms in diverse IoT scenarios. OCUS offers promising applicability in real-world domains such as urban infrastructure management and emergency response, where adaptive and coordinated UAV operations can significantly enhance the delivery of services.

1. Introduction

Unmanned Aerial Vehicles (UAVs) have gained significant popularity due to their diverse applications [1] such as communication [2, 3], disaster management [4], surveillance [5], parcel delivery [6], and agriculture [7]. The usefulness of individual UAVs comes with substantial upfront capital expenditure (CAPEX) and continued operational expenditure (OPEX). Additionally, different applications demand specialized UAVs with varying abilities, such as sensor types, battery capacity, or computational capability. Reducing these increased expenses is essential for making UAV deployment more cost-effective. To address the issues of significant CAPEX and OPEX of UAVs for

single application, Yapp et al. [8] proposed a new architecture as UAV-as-a-Service (UaaS). By leveraging the concept of the traditional cloud computing, UaaS provide an opportunity to the users without the requirement of direct ownership or maintenance of the UAVs by themselves. Similarly, Nait-Abdesselam et al. [9] proposed the D3S framework, consisting of four phases – demand, decision, deployment, and service – to tackle challenges such as UAV selection, optimal deployment, and trajectory planning. In a UaaS platform, UAVs are virtualized and provided as a service, allowing multiple end-users to access UAV functionalities on-demand for multiple applications [10]. UaaS platform optimizes resource utilization and provides substantial

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financial profits. The UaaS platform encourages UAV owners to rent their UAVs to the platform, which allows them to generate revenue from their UAVs. At the same time, end-users also benefit by avoiding the high CAPEX of buying UAVs and the ongoing OPEX of maintenance and management. Challenges like service continuity during UAV transitions and trust based task offloading have been addressed through schemes such as Serv-HU [11] and TOFU [12] for UaaS.

In spite of the financial and operational advantages that the UaaS platform provides, some important challenges still require to address, especially in the efficient management and selection of UAVs. One of the important problems within the UaaS platform is the formation of UAV coalitions. Since each end-user application has unique requirements, such as specific sensor types, flight duration, reputation of UAV, geographical constraints, or energy efficiency, it is important to assemble a group of UAVs that meet these diverse demands efficiently. Regardless, the heterogeneity of UAVs in the platform, which comes from different UAV owners and has varying capabilities, raises complexity. Forming an optimal coalition with this heterogeneous set of UAVs requires addressing multiple objectives, such as — maximizing task success, minimizing energy consumption, and reducing operational costs. An improper coalition formation results in inefficient use of resources, higher energy consumption, increased operational costs, and potential delays in application execution.

In this work, we propose a scheme, OCUS, that forms an optimal coalition of UAVs using a coalition game theory, where each UAV is a player. Coalition game theory provides a mathematical framework where multiple autonomous agents (in our case, UAVs) form groups or coalitions to achieve a common objective that may not be attainable individually [13]. OCUS form coalition to maximize application success while minimizing resource use and operational costs. By leveraging Nash Equilibrium (NE), OCUS guarantees that no UAV is incentivized to deviate from the coalition, leading to stable and efficient collaborations that meet the specific application requirements. The proposed approach optimizes performance, improves task completion rate, reduces energy consumption, and enhances the overall efficiency of the UaaS platform.

1.1. Motivation

Typically, in a UaaS platform, multiple heterogeneous UAVs collaborate to serve an application. One of the crucial challenges in a UaaS platform is the optimal selection of UAVs to form a coalition. Further, each application possesses specific requirements, such as the types of sensors required, how long the UAVs are required to hover in the air, and how much energy they will consume. However, UAVs are resource constraint in nature, having limited resources such as battery, processing power, and storage capacity. An improper coalition of UAVs leads to high operational costs, inefficient resource allocation, and lower service quality. On the other hand, UAVs in the UaaS platform are often owned by different owners, leading to varying objectives, pricing strategies, and operational constraints. Managing cooperation and coordination among such heterogeneous entities becomes non-trivial. Issues such as fairness in task allocation, and incentivization for long-term participation further complicate coalition formation. Additionally, the UaaS platform is new and unique, so the existing coalition formation schemes designed for other domains may not be suitable. Consequently, there is an utmost requirement to design an optimal UAV coalition scheme while focusing on the uniqueness of a UaaS platform. Therefore, we are motivated to develop an optimal coalition formation scheme that maximizes task completion rates while minimizing energy consumption and resource usage costs.

1.2. Contribution

This work proposes an optimal UAV coalition formation scheme, OCUS, using a game-theoretic approach. The proposed scheme, OCUS, forms optimal UAV coalitions that efficiently meet each application's distinct requirements. Our specific *contributions* are as follows:

- First, to ensure effective UAV coalition formation for the UaaS platform, we introduce a utility-based scheme for optimally selecting a set of UAVs to complete specific applications. The proposed approach incorporates a mathematical formulation for UAV-task assignment, where the utility $U(u_i, t_j)$ represents a sensor compatibility, task execution agility, and proximity to the application area.
- Second, we develop a comprehensive UAV owner selection mechanism for the UaaS platform, incorporating key performance metrics to ensure fairness and efficiency in task allocation. This mechanism evaluates UAV owners based on their Reputation Score, $R_S(uo_i)$, Cost Competitiveness Ratio, $C_R(uo_i)$, and Task Allocation Fairness Factor, $T_A(uo_i)$. The Reputation Score is calculated using indicators such as Success Rate, $\rho(uo_i)$, On-Time Delivery, $\infty(uo_i)$, and Average Satisfaction, $\mu_s(uo_i)$, reflecting service quality. Based on these metrics, a selection score is assigned to each UAV owner, enabling optimal selection with a balance of cost-effectiveness, reliability, and fair task distribution.
- Third, we introduce a payoff-driven coalition formation scheme, OCUS, designed to ensure stability and maximize utility in UAV coalitions within the UaaS platform. OCUS uses a payoff function, $\pi_i(\lambda_j)$, for each UAV u_i in coalition λ_j , considering key metrics such as Utility ($U(u_i, t_z)$), Reputation Score ($R_S(uo_i)$), and Task Allocation Fairness ($T_A(uo_i)$), along with cost and equilibrium constraints. OCUS optimizes key performance objectives in the UaaS platform, including high task completion rate, energy efficiency, coalition stability, and fairness in terms of participation among UAV owners. Using the Nash Equilibrium, we stabilize the coalition, while ensuring that UAVs are unable to improve their payoffs by changing coalitions. We also compare the performance of OCUS with state-of-the-art approaches – FedBarg, MSCGTO, and a Decentralized scheme – and provide a detailed performance analysis demonstrating the advantages of our scheme over existing solutions.

2. Related work

This section discusses existing work on multiple service-oriented architectures (SOA) and various coalition formation strategies. To address security concerns in Software-as-a-Service (SaaS) for outsourcing and enabling multi-tenancy, Yassin et al. [14] introduced Multi-Tenant Intrusion Detection as a Service (MTIDaaS). The proposed framework, MTIDaaS, allows service providers to incorporate security measures seamlessly, offering Security-as-a-Service (SecaaS) to providers and tenants. MTIDaaS is designed to provide high portability, flexibility, and cost-effectiveness, providing robust security without compromising efficiency. Further, to ensure consistent data access at a reasonable cost, Al-Zahrani et al. [15] proposed a blockchain-based subscription model for Data-as-a-Service (DaaS). In DaaS, users subscribe to Data Providers (DP) for a fixed period and pay as per plan. DaaS permits DPs to gain recurring revenue over time, proposing a higher profit margin than one-time data sales. On the other hand, to remove the limitation of stationarity of sensor nodes in the sensor service provisioning system, Roy et al. [16] proposed the Mobile Sensor-Cloud (MSC) architecture. An MSC architecture is based on mobile Sensors-as-a-Service (mSe-aaS) and leverages the mobility of devices with sensors such as accelerometers, gyroscopes, GPS, and vibration detectors. Further, in another work [17], the authors presented a pricing model called PRIME for mSe-aaS. On the other hand, Roy et al. [18] introduced Safety-as-a-Service (Safe-aaS) for the road transportation sector, offering dynamic and personalized safety solutions to registered users based on their specific requirements. The service operates within a service-oriented architecture involving sensor owners, vehicle owners, safety providers, and end-users. Further, in Safe-aaS, multiple challenges were addressed, such as service handoff method [19], termed as Safe-Passé and Region-Based Pricing Scheme for Provisioning Safe-aaS [20]. Similarly, Hasić et al. [21] proposed a Decision-as-a-Service (DaaS) model within a layered SOA, where decisions are automated

services that processes call on-demand. The approach is explained using formalized Decision Model and Notation (DMN) constructs and relevant layers. For zero touch and zero trust between multiple stakeholders, Valero et al. [22] proposed Trust-as-a-Service (TaaS). Moreover, recent research highlights the importance of intelligent routing in UAV networks to improve communication efficiency and energy use. For example, Shuffled Frog Leaping Algorithm (SFLA)-based routing has been proposed to overcome limitations like high latency, low adaptability, and energy waste in traditional methods [23]. At the same time, secure coordination in UAV systems is gaining traction. Heidari et al. [24] proposed a blockchain-enabled radial basis function (RBF) neural network for the Internet of Drones, helping improve decentralized decision-making and protect data integrity in dynamic aerial networks.

With the advancement of SOA-based architecture, there is significant improvement in resource management and data security. Hadjres et al. [25] proposed a coalition formation approach based on social games called the service level agreement (SLA)-Aware Cloud Coalition Formation algorithm (S-ACCF) to handle peak-time client demand for cloud services. S-ACCF identifies the most effective coalition of cloud providers to fulfill requests while meeting client SLA requirements. On the other hand, Kakkar et al. [26] introduced an optimal data pricing model for ride-sharing services, facilitating transactions between vehicle owners and customers by utilizing blockchain technology and coalition game theory. Coalition game theory used in this to maximize the payoffs for both vehicle owners and customers by encouraging coalition formation. Similarly, to optimize resource allocation in a dynamic multiservice system, Lin et al. [27] introduced the Multistage Coalition Game Task Offloading (MSCGTO) framework, designed to handle scalability, dynamic conditions, and diverse resources. MSCGTO framework includes two distributed algorithms that work together to maximize the service provider's profit and minimize the end-to-end delay for service subscribers, balancing cost-effectiveness with acceptable user latency. In a similar effort, Chakraborty et al. [28] proposed a scheme called FedBarg, which combines coalition formation games and utility theory to achieve optimal resource allocation for wireless edge networks. FedBarg approach aims to find the most effective solutions for resource distribution. On the other hand, Pinto et al. [29] introduced a coalition game theory model to help electricity consumers form coalitions and trade their flexibility in the electricity market. This model focuses on grouping consumers to maximize benefits without direct utility transfers. To share resource between service providers and edge cloud, Zafari et al. [30] proposed a Cooperative Game Theory (CGT)-based framework for resource sharing among edge cloud providers, prioritizing native applications and allocating surplus resources to others. Two algorithms, Game-theoretic Pareto Optimal Allocation (GPOA) and Polyandrous-Polygamous Matching Pareto Optimal Allocation (PPMPOA), ensure stable and Pareto-optimal resource allocations. Whereas Gupta et al. [31] offers a detailed review of the potential advantages of leveraging game theory (GT) in wireless communication (WC). To address the resource constraints in computing, storage, and communication at mobile edge computing (MEC) servers, Wang et al. [32] proposed a hierarchical resource management framework that enables collaborative optimization across multiple domains. Their approach jointly considers task offloading, scheduling, and caching decisions with the goal of minimizing overall system delay. The problem is decomposed into two parts: local resource allocation within individual MEC servers and load balancing across multiple MEC nodes to enhance system performance. Further, Heidari et al. [33] survey offers a comprehensive review of evolving botnet threats, command and control mechanisms, and evasion strategies. On the other hand, to address real-time resource allocation for service requests from road vehicles, the authors in [34] proposed HARMONIC – a game theory-based coalition model designed to improve resource utilization and maintain a dynamic balance across multiple vehicular clouds.

Synthesis: Existing research shows much advancement in SOA-based architecture and coalition formation across various domains. The existing solutions such as MTIDaaS and blockchain-based DaaS address

security and data-sharing needs, while frameworks like MSC and SafeaaS improve flexibility and collaboration by enabling mobile sensing and safety services. However, despite these advancements, there is no research on coalition formation within the UAV-as-a-Service (UaaS) platform. This gap highlights the requirement for novel approaches to coalition formation that consider the unique requirements and operational complexities of UaaS, which significantly enhance efficiency and adaptability in UAV service platforms. Unlike MSCGTO, which focuses solely on latency optimization, or S-ACCF, which models cloud provider coalitions, OCUS supports energy-constrained UAV coalitions with heterogeneous capabilities and owners. It integrates a payoff model based on sensor-task alignment, proximity, and reputation, while ensuring long-term coalition stability through a game-theoretic formulation. This enables OCUS to provide a more holistic and practical coalition formation mechanism for real-world UaaS platform.

3. Problem description

3.1. Problem scenario

In a UaaS platform, the key challenge lies in efficiently forming a coalition of UAVs to execute complex applications efficiently. Multiple actors – UAV owners, service providers, and end-users – are involved in this platform. UAV owners lease their UAVs to the platform, managed by service providers who is responsible for coordinating the distribution and selection of UAVs to meet the end-user's application requirements. Each application submitted by an end-user demands specific sensor capabilities, which must be fulfilled by a combination of UAVs, considering factors such as flight duration, payload capacity, communication range, real-time processing and computation, agility, speed, and geographical coverage. The primary issue is that UAVs differ significantly in terms of their capabilities. This heterogeneity is because the UAVs are sourced from different owners on the platform. As a result, these UAVs vary in sensor types, battery life, flight range, and energy consumption. Forming an optimal coalition, therefore, requires selecting UAVs that complement each other to meet the specific demands of each task.

The geographical constraints such as proximity to task location introduce another layer of complexity, as UAVs closer to the application location may be prioritized for selection. However, in cases where proximity is not ideal, the platform may need to coordinate dynamic handoffs between UAVs, further complicating the task completion process. Redundancy must also be minimized, as selecting more UAVs than necessary or creating inefficient coalitions wastes resources and drives up costs. So, the core problem is figuring out how to form an optimal UAV coalition that meets the application-specific requirements, makes the best use of energy, keeps costs down, and ensures resources are used efficiently, all within the UaaS platform.

3.2. Mathematical model

A UaaS platform primarily consists of three actors – end-user, the service provider, and the UAV owner. Let $UO = \{uo_1, uo_2, \dots, uo_m\}$ be the set of UAV owners, where m represents the total UAV owners in the UaaS platform. Each UAV owner rents one or more UAVs to the platform. Let $U = \{u_1, u_2, \dots, u_n\}$ be the set of available UAVs with service provider in the UaaS platform. The total number of UAVs, n , is the sum of the UAVs provided by all owners. Therefore, we represent the total number of UAVs present in the systems as follows:

$$U = \sum_{i=1}^m u_i \quad (1)$$

The set of sensors on UAV u_j is expressed as $S_j = \{s_j^1, s_j^2, \dots, s_j^p\}$, where, s_j^k is the k th sensor hosted on UAV u_j and p is the total number of sensors on UAV u_j . A represents an application that requires to be executed

on the UaaS platform. Let $CP^b = \{CP_1^b, CP_2^b, \dots, CP_m^b\}$ represents the set of base service charges by each UAV owner for completing a single application. Additionally, let $CP^u = \{CP_1^u, CP_2^u, \dots, CP_m^u\}$ denotes the per unit charge applied by each UAV owner for the services provided.

Let a UAV owner deploy k UAVs to complete a task, where each UAV requires τ units of time to finish its assigned portion. Let t_j^{ser} denotes the total time taken by the UAVs of owner uo_j to complete the task. Therefore,

$$t_j^{ser} = \sum_{i=1}^k \tau \quad (2)$$

The UAV owner charges a standard, uniform rate for all UAVs in their fleet. The total cost incurred by the owner to complete an application includes a base service charge (CP_j^b) along with charges per unit (CP_j^u) for each UAV deployed. Let CP_j^{tot} denotes the charged price by a UAV owner, uo_j for completing task that takes a total time of t_j^{ser} .

$$CP_j^{tot} = CP_j^b + (CP_j^u \times t_j^{ser}) \quad (3)$$

The average total price charged by all UAV owners for completing a task is represented by CP_{avg} . It is defined as the mean of CP_j^{tot} for all uo_j .

$$CP_{avg} = \frac{\sum_{j=1}^m CP_j^{tot}}{m} \quad (4)$$

The location of the application area L_A is represented with 3-tuple – longitude, latitude, and altitude, such that: $L_A = \langle Lon, Lat, Alt \rangle$, where Lon , Lat , and Alt represent longitude, latitude, and altitude, respectively. Let $T = \{t_1, t_2, \dots, t_k\}$ represent the set of tasks for the application A . Each task, t_j , requires a specific set of sensors for its execution. Let $S_{req,j}$ be the set of required sensors for task t_j . The total set of sensors required for the entire application A is the union of the sensors required for all tasks.

$$S_{req,A} = \bigcup_{j=1}^k S_{req,j} \quad (5)$$

where $S_{req,A}$ is the total set of sensors required for the application A . Let $\sigma(uo_i)$ be the total number of tasks assigned to all the UAVs of owner uo_i .

As time is a essential factor in providing efficient services, we include different time factor in our work.

Takeoff time ($t_{takeoff}$) refers to the time required for a UAV to ascend from a stationary ground position to its designated operational altitude.

$$t_{takeoff} = \left(\frac{H_{op}}{v_{asc}} \right) + \left(\frac{W \times H_{op}}{P_{thrust}} \right) + f(W, A) \quad (6)$$

where H_{op} , v_{asc} , W , and P_{thrust} , are operating altitude, ascent rate, UAV weight, and thrust power, respectively. Further, $f(W, A)$ is a function accounting for weight W and weather effects such as wind speed and temperature.

Flight time (t_{flight}) of a UAV is defined as the total duration from the completion of ascent phase until the initiation of descent phase.

$$t_{flight} = \left(\frac{D}{v_{cruise}} \right) + \left(\frac{W \times D}{P_{prop}} \right) + g(W, A, F) \quad (7)$$

where D , v_{cruise} , P_{prop} , and $g(W, A, F)$ are distance to the task location, cruising speed of the UAV, propulsion power, and function accounting for the weight (W), air density (A), and weight effects during flight (F), respectively.

Landing time ($t_{landing}$) refers to the duration required for a UAV to descend from its operational altitude to its final stationary position on

the ground. This phase begins once the UAV initiates its descent and continues until it safely completes its landing.

$$t_{landing} = \left(\frac{H_{op}}{v_{desc}} \right) + \left(\frac{W \times H_{op}}{P_{thrust}} \right) + h(W, A) \quad (8)$$

where H_{op} , v_{desc} , and $h(W, A)$ are operational altitude, descent speed of the UAV, and a function accounting for weight (W) and weather effects (A) such as wind speed and temperature.

The time required by UAV u_i to complete task t_j , denoted as Y_{ij} , such that:

$$Y_{ij} = (t_{takeoff} + t_{flight} + t_{landing}) \quad (9)$$

As energy is an essential resource for UAV, we consider energy capacity, $E_{cap}(u_i)$, as a parameter for the coalition formation. To better approximate UAV energy limitations in real-world long-term deployments, we incorporate battery degradation into the energy capacity model. Specifically, we model the effective capacity of UAV batteries using Peukert's Law [35], which captures the relationship between discharge rate and usable energy capacity over time. $E_{cap}(u_i)$ is the maximum energy that the UAV have while in operation.

$$E_{cap}(u_i) = \frac{E_{nom}(u_i)}{(I(u_i))^{k-1}} \quad (10)$$

where $E_{nom}(u_i)$ is its nominal battery capacity, $I(u_i)$ is the average discharge current during operation, and k is the Peukert exponent, typically in the range [1.1, 1.3].

The value of $E_{cap}(u_i)$ must be greater than the energy needed for payload activities, onboard computations and propulsion. Therefore,

$$E_{cap}(u_i) \geq \sum_{j \in T_i} (E_{prop}(u_i, t_j, W, A) + E_{comp}(u_i, t_j) + E_{pay}(u_i, t_j)) \quad (11)$$

where $E_{prop}(u_i, t_j, W, A)$ denotes propulsion energy required for u_i to complete task t_j , factoring in its weight W , and weather effects A , which include wind speed, temperature, and air pressure, $E_{comp}(u_i, t_j)$ is computation energy required for real-time processing onboard, and $E_{pay}(u_i, t_j)$ is energy required for payload operations.

Let X be a matrix representing the assignment of UAVs to tasks in the UaaS platform. Each entry x_{ij} in X denotes the assignment status of UAV u_i to task t_j , which provides a structured way to capture UAV-task allocations. The entries of matrix X are defined as follows:

$$x_{ij} = \begin{cases} 1, & \text{if UAV } u_i \text{ is assigned to task } t_j \\ 0, & \text{otherwise} \end{cases} \quad (12)$$

The energy consumption of UAV u_i during task t_j , is represented by z_{ij} . The value of z_{ij} depends on different factors such as complexity of task, task duration, and sensors present in UAV. If we assume that an UAV does total e_i tasks, this follows a constraint that:

$$\forall u_i \in U, \quad \sum_{j=1}^{e_i} z_{ij} \leq E_{cap}(u_i) \quad (13)$$

We derive a utility function, $U(u_i, t_j)$, to select a suitable UAV for a task, t_j . $U(u_i, t_j)$ consists SensorMatch, Task Execution Agility, and Effective Proximity Factor, discussed as follows:

SensorMatch ($SM(u_i, t_j)$) measures how well u_i satisfies sensor requirement for t_j , represented as:

$$SM(u_i, t_j) = \frac{1}{|S_{req,j}|} \sum_{s \in S_{req,j}} \delta(s \in S_i) \times \omega_s(u_i) \quad (14)$$

where $S_{req,j}$ is the number of sensors required for t_j , $\delta(s \in S_i)$ is an indicator function that returns 1 if sensor s is available in UAV u_i , and 0 otherwise. The term $\omega_s(u_i)$ is a sensor quality coefficient that accounts for both sensor accuracy and calibration reliability of sensor s on UAV u_i , normalized to [0,1]. This formulation ensures that UAVs with more precise and better-calibrated sensors are prioritized during coalition formation.

Task Execution Agility ($TEA(u_i, t_j)$) is the reciprocal of Y_{ij} which is derived using Eq. (9), which determines the ability of UAV u_i to complete task t_j in a given time based on its operational speed.

$$TEA(u_i, t_j) = \frac{1}{Y_{ij}} \quad (15)$$

Effective Proximity Factor ($EPF(u_i, L_A)$) is the reciprocal of the physical distance between UAV u_i and the location of task t_j ($D(u_i, L_A)$), such that a UAV far away from task location are assigned a lower weight.

$$EPF(u_i, L_A) = \frac{1}{D(u_i, L_A)} \quad (16)$$

To capture signal propagation effects in dense urban environments, we extend the EPF in Eq. (16) to an Urban EPF (EPF_{urban}) by integrating the Okumura-Hata (OH) radio propagation model [36].

First, the path loss L_p (dB) between UAV u_i and the task location is computed using the OH model for urban areas:

$$L_p(\text{dB}) = 69.55 + 26.16 \log_{10}(f) - 13.82 \log_{10}(h_b) - a(h_r) + [44.9 - 6.55 \log_{10}(h_b)] \log_{10}(d) \quad (17)$$

where f is the signal frequency in MHz, h_b is the height of the UAV (transmitter) in meters, h_r is the height of the task location (receiver) in meters, d is the horizontal distance in kilometers, and $a(h_r)$ is the antenna height correction factor for the receiver.

This path loss is then converted to a linear attenuation factor A_{urban} using $A_{urban} = 10^{-L_p/10}$. The new EPF_{urban} is then defined by scaling the original EPF (from Eq. (16)) with this attenuation factor:

$$EPF_{urban}(u_i, L_A) = EPF(u_i, L_A) \times A_{urban} \quad (18)$$

In urban deployment scenarios, Eq. (18) replaces the standard EPF in the overall utility calculation (Eq. (19)). This ensures that the utility of a UAV accurately reflects the quality of its communication link, penalizing positions with high signal attenuation.

Furthermore, we can enhance A_{urban} by including small-scale fading. The enhanced attenuation factor would be computed as $A_{urban_enhanced} = A_{urban} \times F_{fading}$, where F_{fading} represents the small-scale fading component that can be modeled as either Rayleigh fading (in the absence of a dominant line-of-sight path), given by $F_{rayleigh} = |h|^2$, where $h \sim \mathcal{CN}(0, \sigma^2)$ is a complex Gaussian random variable, or Rician fading, given by $F_{rician} = |h_{LOS} + h_{multipath}|^2$, where h_{LOS} represents the direct path component and $h_{multipath}$ represents the scattered multipath components. To make our model more realistic, we propose using the UAV altitude and urban density metrics to adaptively select between Rayleigh and Rician fading: at low altitude (<15 m), fading is predominantly Rayleigh due to heavy obstruction, whereas at higher altitude (≥ 15 m), Rician fading is assumed with the K -factor varying based on altitude and distance.

Property 1. The utility function $U(u_i, t_j)$ is linear to its contributing factors: $SM(u_i, t_j)$, $TEA(u_i, t_j)$, and $EPF(u_i, L_A)$.

The utility function $U(u_i, t_j)$ is designed to select the UAV that is best suited for task t_j based on three key factors: sensor match $SM(u_i, t_j)$, task execution agility $TEA(u_i, t_j)$, and effective proximity factor $EPF(u_i, L_A)$. Property 1 ensures that the utility is computed as a scaled sum of these independent factors, where each component contributes proportionally based on its respective scaling constant. Using Eqs. (14), (15) and (16), we define utility function as:

$$U(u_i, t_j) = \alpha_1 \times SM(u_i, t_j) + \alpha_2 \times TEA(u_i, t_j) + \alpha_3 \times EPF(u_i, L_A) \quad (19)$$

where α_1 is a dimensionless constant of proportionality, α_2 is scaling constants with units of time, and α_3 is scaling constants with units of distance.

Table 1
Key notations.

Notation	Description
U	Set of all UAVs
uo_i	i th UAV owner
S_i	Set of sensors available on UAV u_i
$S_{req,j}$	Required set of sensors for task t_j
L_A	Location of application
Y_{ij}	Time required by u_i to complete task t_j
$U(u_i, t_j)$	Utility of UAV u_i for task t_j
$SM(u_i, t_j)$	Sensor Match between UAV u_i and task t_j
$TEA(u_i, t_j)$	Task Execution Agility of UAV u_i for task t_j
$EPF(u_i, L_A)$	Effective Proximity Factor of UAV u_i to application
$\pi_i(\lambda_j)$	Payoff of UAV u_i in coalition λ_j
$R_S(uo_i)$	Reputation score of uo_i
$C_R(uo_i)$	Cost competitiveness ratio of uo_i
$T_A(uo_i)$	Task Allocation Fairness factor of uo_i
z_{ij}	Energy consumption of UAV u_i for task t_j
$E_{cap}(u_i)$	Energy capacity of UAV u_i
$v_{asc}, v_{desc}, v_{cruise}$	Ascent, descent, and cruise velocity of UAV
P_{thrust}, P_{prop}	Power required for takeoff and propulsion
$\rho(uo_i)$	Success rate of uo_i
$O(uo_i)$	On-time delivery metric of uo_i
$\mu_s(uo_i)$	Average satisfaction score of uo_i

Property 2. The selected UAVs must host all the types of sensors that are required to perform a task for an application.

Property 2 increases the likelihood that the application will be completed by ensuring that each UAV is best equipped to meet the sensor dependent requirements of its assigned application. Therefore,

$$\forall t_j \in T, \bigcup_{i=1}^n (x_{ij} \times S_i) \supseteq S_i \cap S_{req,j} \quad (20)$$

where T is total number of task in an application A .

Property 3. The time required to complete the task must be less than a predefined maximum allowable time, $t_{max,A}$.

This essential restriction acts as a buffer against application failures or delays that may occur when exceeding the allotted time limits.

$$\sum_{i=1}^n \sum_{j=1}^k x_{ij} Y_{ij} \leq t_{max,A} \quad (21)$$

To identify UAV owners offering cost-effective pricing to service providers, we define a new parameter called the cost competitiveness ratio, using Eqs. (3) and (4).

Definition 1. The Cost competitiveness ratio ($C_R(uo_j)$) for a UAV owner uo_j is the ratio of the price quoted by uo_j to the average price offered by other owners for similar tasks.

$$C_R(uo_j) = \frac{CP_j^{tot}}{CP_{avg}} \quad \forall uo_j \text{ in } UO \quad (22)$$

In Eq. (22), a ratio less than 1 indicates that the UAV owner charges a price below the market average, making it a cost-competitive option. Conversely, a ratio greater than 1 implies that the UAV owner's price exceeds the market average, suggesting it is less cost-competitive.

Definition 2. The Task Allocation Fairness Factor $T_A(uo_i)$ ensures fairness in terms of distribution of tasks among new and old UAV owners.

It prioritizes UAV owners that have been assigned fewer or no tasks in previous application assignments, promoting fairness in task

allocation. $T_A(uo_i)$ helps balance the workload across all available UAV owners on UaaS platform.

$$T_A(uo_i) = \frac{1}{\sigma(uo_i) + 1} \quad (23)$$

We add 1 in the denominator of Eq. (23) to ensures baseline fairness for UAV owners, especially those with minimal or no tasks assigned. It prevents UAV owners with very few tasks from being penalized disproportionately.

The Task Completion Status for task t_j , denoted as $C(t_j)$ is a binary variable indicating whether the task was completed successfully.

$$C(t_j) = \begin{cases} 1, & \text{if task } t_j \text{ was completed successfully} \\ 0, & \text{otherwise} \end{cases} \quad (24)$$

To better capture recent performance trends, we redefine the success rate for UAV owner uo_i , denoted as $\rho(uo_i)$, using an exponentially weighted formulation. This approach emphasizes recent task completions while progressively discounting older ones. Specifically, $\rho(uo_i)$ is defined as:

$$\rho(uo_i) = \frac{\sum_{j=1}^{\sigma(uo_i)} C(t_j) \times e^{-\Delta t_j / \tau}}{\sum_{j=1}^{\sigma(uo_i)} e^{-\Delta t_j / \tau}} \quad (25)$$

where, $C(t_j)$ is a binary indicator representing whether task t_j was successfully completed ($C(t_j) = 1$ if successful, 0 otherwise), and Δt_j denotes the time elapsed since task t_j was completed. The parameter τ is the decay constant that governs the rate of exponential attenuation. A smaller τ places more weight on recent tasks, allowing the system to adapt more quickly to behavioral changes in UAV owner performance.

Definition 3. The Delivery Success Indicator for task t_j , denoted as $D_s(t_j)$, is a binary variable that represents whether the delivery for task t_j was completed on time.

$$D_s(t_j) = \begin{cases} 1, & \text{if task } t_j \text{ was completed on time} \\ 0, & \text{otherwise} \end{cases} \quad (26)$$

To reflect the timeliness of recent task completions, we define the decayed On-Time Delivery for uo_i , denoted as $O(uo_i)$, as the exponentially weighted ratio of tasks completed on time:

$$O(uo_i) = \frac{\sum_{j=1}^{\sigma(uo_i)} D_s(t_j) \times e^{-\Delta t_j / \tau}}{\sum_{j=1}^{\sigma(uo_i)} e^{-\Delta t_j / \tau}} \quad (27)$$

Definition 4. The Satisfaction Score given for t_j , denoted as $Q(t_j)$, is a value between 0 and 1 that indicates the end-user satisfaction with the completion of A .

The Satisfaction score assigned to each task within the application is the same as overall score given to the application, reflecting a uniform level of satisfaction across all tasks. The rating is determined based on feedback collected from the end-user and is normalized between 0 and 1, where 0 indicates complete dissatisfaction and 1 indicates maximum satisfaction.

To better reflect evolving end-user experiences, we define the decayed average satisfaction for UAV owner uo_i , denoted as $\mu_s(uo_i)$, using exponential time-discounting:

$$\mu_s(uo_i) = \frac{\sum_{j=1}^{\sigma(uo_i)} Q(t_j) \times e^{-\Delta t_j / \tau}}{\sum_{j=1}^{\sigma(uo_i)} e^{-\Delta t_j / \tau}} \quad (28)$$

This decayed formulation ensures that recent user feedback carries greater weight, enabling the system to adapt more responsively to changes in UAV owner performance.

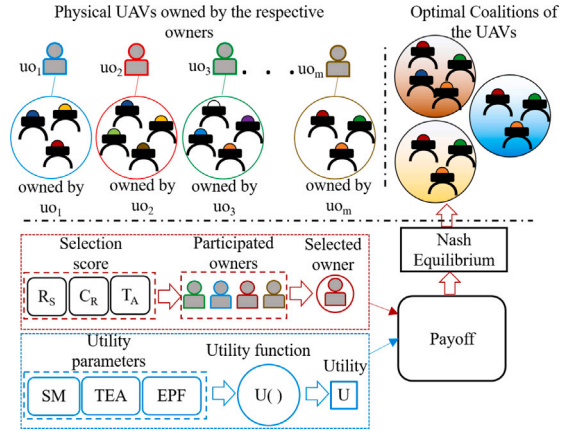


Fig. 1. OCUS architecture.

Definition 5. The Reputation Score, denoted as $R_S(uo_i)$, evaluates the reputation of a UAV owner based on their past performance.

We compute reputation score using Eqs. (25), (27), and (28) which collectively contributes to the UAV owner's overall standing, reflecting the consistency, quality, and trustworthiness of their services. We decouple technical performance from user experience as follows:

$$R_T(uo_i) = \beta_1 \times \rho(uo_i) + \beta_2 \times O(uo_i) \quad (29a)$$

$$R_S^{\text{subj}}(uo_i) = \beta_3 \times \mu_s(uo_i) \quad (29b)$$

$$R_S(uo_i) = \gamma_1 \times R_T(uo_i) + \gamma_2 \times R_S^{\text{subj}}(uo_i) \quad (29c)$$

where $R_T(uo_i)$ is technical performance, $R_S^{\text{subj}}(uo_i)$ is user experience, β_1 , β_2 and β_3 are dimensionless constant of proportionality, $\gamma_1 + \gamma_2 = 1$, and can be tuned by the platform. This decomposition improves interpretability and flexibility in coalition formation.

Further, the selection score aggregates three key parameters: reputation score $R_S(uo_i)$, cost competitiveness ratio $C_R(uo_i)$, and task allocation fairness factor $T_A(uo_i)$. Each term is scaled by its corresponding weight α_1 , α_2 , and α_3 , reflecting its relative importance in evaluating a UAV owner's suitability. A higher selection score increases the likelihood of a UAV owner's UAVs being chosen for coalition formation.

$$\text{SelectionScore} = \alpha_1 \times R_S(uo_i) - \alpha_2 \times C_R(uo_i) + \alpha_3 \times T_A(uo_i) \quad (30)$$

where α_1 , α_2 and α_3 are dimensionless constant of proportionality. The higher the selection score, the greater the likelihood of the UAV owner being selected.

Fig. 1 illustrates the OCUS architecture, which optimizes coalition formation among UAVs within the UaaS platform. The architecture begins with UAV owners deploying their UAVs to the platform, making them available for various IoT-based applications requested by end-users. Upon receiving an application, the OCUS scheme strategically forms optimal coalitions of UAVs based on multiple selection scores and utility parameters.

4. OCUS: The solution approach

In Nash Equilibrium, payoff refers to the quantified benefit a UAV gets based on their chosen strategy and coalition, considering factors like utility, reputation score, task allocation fairness, cost competitiveness ratio, and energy. Payoff determines the UAV's incentive to remain in or deviate from the coalition, with equilibrium achieved when no UAV can unilaterally improve their payoff. The utility ($U(u_i, t_j)$) reflects how well UAV u_i is suited for task t_j , considering factors such as sensor

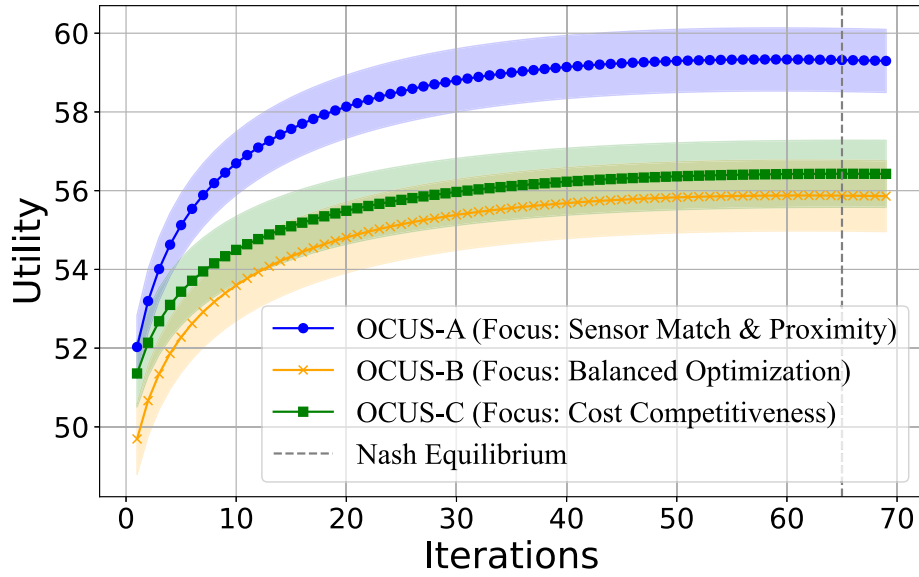


Fig. 2. Nash equilibrium achieved via multiple strategies.

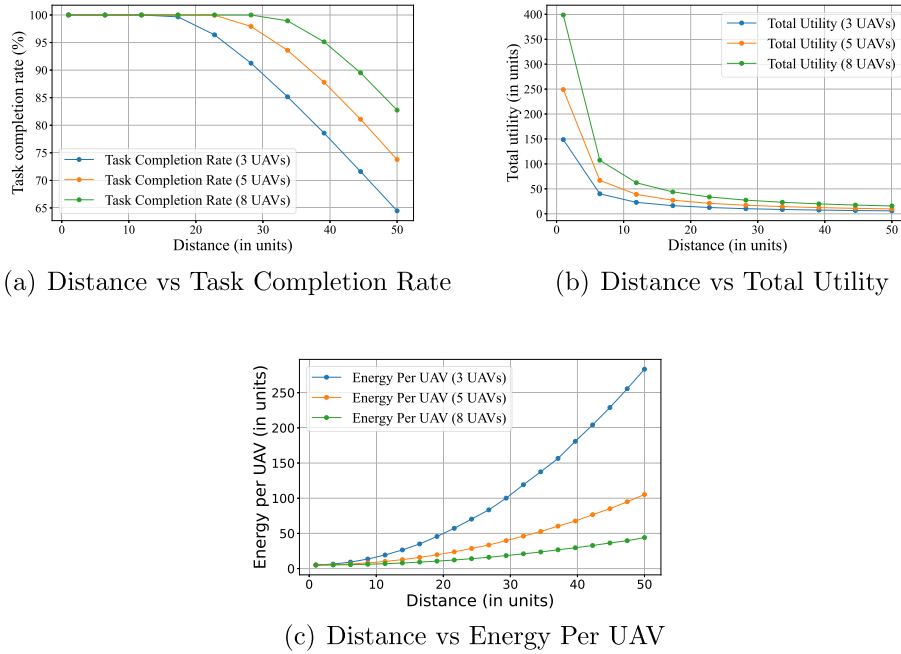


Fig. 3. Comparison of various distance-related parameters for varying number of UAVs.

compatibility, agility, and proximity. Payoff ($\pi_i(\lambda_j)$) extends this by incorporating coalition-level aspects such as owner reputation, fairness, cost, and energy use. While utility aids in UAV-task selection, payoff supports stable coalition formation.

The payoff function for UAV u_i in coalition λ_j is denoted as $\pi_i(\lambda_j)$.

$$\pi_i(\lambda_j) = \alpha_1 \times U(u_i, t_k) + \alpha_2 \times R_S(uo_i) + \alpha_3 \times T_A(uo_i) - \alpha_4 \times C_R(uo_i) - \alpha_5 \times z_{ik} \quad (31)$$

where z_{ik} is energy consumed by UAV u_i to complete task t_k and $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ and α_5 are scaling constant of dimensionless, $R_S(uo_i)$ captures a UAV owner's reputation based on success rate, on-time delivery, and satisfaction (Eq. (29)), while $T_A(uo_i)$ reflects fairness by favoring underutilized owners (Eq. (23)). Together, they help balance payoff

with past performance and equitable task distribution. Nash Equilibrium is achieved through an iterative convergence process rather than through direct computation. Each UAV evaluates its payoff using Eq. (31) and determines whether joining a different coalition would improve its outcome. At equilibrium, each UAV u_i should belong to a coalition in which its marginal payoff is non-positive. This implies that the UAV either gains no additional benefit or incurs a loss by switching the coalition, that ensures stability within the coalition. When a new task or UAV enters, the re-evaluation of utility and payoff leads to a bounded number of reassignments due to the concavity of payoff and diminishing gains from deviation — thus maintaining local coalition stability.

$$\frac{\partial \pi_i(\lambda_j)}{\partial \lambda_j} \leq 0 \quad (32)$$

Eq. (32) represents a necessary condition for equilibrium. A marginal payoff of zero indicates that the UAV is maximizing its payoff within the current coalition, while a marginal payoff less than zero implies that the UAV cannot improve its outcome by switching to a different coalition. In a Nash Equilibrium, the payoff function must also fulfill the second-order condition for maximization. Specifically, this requires that the second derivative of the payoff with respect to the coalition choice is negative, ensuring that the payoff reaches a local maxima within the coalition structure.

$$\frac{\partial^2 \pi_i(\lambda_j)}{\partial \lambda_j^2} < 0 \quad (33)$$

Eq. (33) ensures that the payoff function is concave at the equilibrium point, signifying that UAV u_i is at a local maxima in terms of its payoff. Consequently, u_i cannot achieve a higher payoff by switching coalitions, reinforcing the stability of the equilibrium. In a Nash Equilibrium, all UAVs must be positioned within coalitions where no individual UAV improve its payoff by unilaterally deviating from its current coalition. The total payoff function for all UAVs is expressed as:

$$\sum_{j=1}^n \pi_i(\lambda_j) \quad (34)$$

Theorem 1. *The total payoff across all UAVs is maximized when each UAV's coalition is fixed.*

Proof. To prove Nash Equilibrium globally, we need to verify that for all UAVs u_i , the conditions outlined in Eqs. (32) and (33) are satisfied.

We define Aggregate Benefit denoted as, B_{agg} , which aggregates the payoff of all UAVs within the coalition. It represents the collective efficiency or overall benefit of the coalition structure. B_{agg} is expressed as the sum of individual payoffs of each UAV.

$$B_{agg} = \sum_{j=1}^n \pi_i(\lambda_j) \quad (35)$$

The objective is to maximize aggregate benefit by selecting coalitions that optimize total utility across all UAVs. Optimization must be performed while adhering to constraints on cost and energy consumption, ensuring an efficient and sustainable coalition structure.

For Nash Equilibrium, the first-order optimality condition requires that no UAV unilaterally switch coalitions to increase its payoff. Mathematically, this implies that for each UAV u_i , the derivative of its pay-off with respect to its coalition choice must be negative. This condition ensures that each UAV is in its locally optimal coalition, where any unilateral deviation would not give a higher payoff.

$$\frac{\partial^2 \pi_i(\lambda_j)}{\partial \lambda_j^2} < 0 \quad \forall u_i \quad (36)$$

$$\forall \lambda_j, \quad \text{if } \exists \hat{\lambda}_j \text{ such that } \pi_i(\hat{\lambda}_j) > \pi_i(\lambda_j), \quad (37)$$

$$\text{then } \pi_k(\hat{\lambda}_j) < \pi_k(\lambda_k) \quad \text{for some } k \neq i$$

where $\hat{\lambda}_j$ is the selected coalition. Eq. (37) implies that any unilateral attempt by one UAV to deviate and improve its own payoff will lead to a decreased payoff for at least one other UAV. Consequently, the equilibrium configuration is Pareto optimal, as no UAV improve its payoff without negatively impacting another. $\square$

To enhance the operational robustness of OCUS against mid-task failures (e.g., energy depletion), we introduce a Real-Time Coalition Repair Protocol. When an active UAV fails, this protocol is triggered to identify an optimal replacement and minimize service disruption. The selection of a backup UAV is guided by the principle of marginal payoff gain ($\Delta\pi_i$), consistent with the core game-theoretic scheme of our work (Eq. (32)). For each candidate UAV u_k in the available pool

Algorithm 1 OCUS with Real-Time Coalition Repair

Input:

- $U = \{u_1, u_2, \dots, u_n\}$: Set of UAVs
- $T = \{t_1, t_2, \dots, t_z\}$: Set of tasks
- $\Lambda = \{\lambda_1, \lambda_2, \dots, \lambda_z\}$: Initial coalition

Output: Final, stable coalition set Λ'

// Phase 1: Initial Coalition Formation

```

1: Initialize coalition  $\lambda_j$  for each task  $t_j$ 
2: Set convergence  $\leftarrow$  false
3: while not converged do
4:   convergence  $\leftarrow$  true
5:   for each UAV  $u_i \in U$  do
6:     Set current coalition  $\lambda_{curr}$  containing  $u_i$ 
7:     Call Algorithm 2 to find  $(\lambda_{best}, \pi_{best})$  for  $u_i$ 
8:     if  $\pi_{best} > \pi_i(\lambda_{curr})$  then
9:       Remove  $u_i$  from  $\lambda_{curr}$  and add to  $\lambda_{best}$ 
10:      convergence  $\leftarrow$  false
11:    end if
12:  end for
13: end while
// Phase 2: Real-Time Monitoring and Repair
14: while tasks are in progress do
15:   for each active UAV  $u_i$  in coalition  $\lambda_j$  do
16:     if  $E_{cap}(u_i)$  depletes mid-task then
17:        $U_{avail} \leftarrow$  Get available UAVs
18:        $u_{backup} \leftarrow \arg \max_{u_k \in U_{avail}} (\Delta\pi_k)$ 
19:       Update  $\lambda_j \leftarrow \lambda_j \setminus \{u_i\} \cup \{u_{backup}\}$ 
20:     end if
21:   end for
22: end while
23: return  $\Lambda'$ 

```

U_{avail} , the system calculates the potential payoff gain from joining the distressed coalition. The UAV that offers the maximum positive gain is selected, ensuring that the replacement is not merely a substitute, but the best possible contributor to the coalition's stability and performance. The efficiency of this protocol is quantified by its Recovery Latency ($T_{recovery}$), defined as the sum of the time taken for failure detection (T_{detect}), backup selection (T_{select}), and deployment (T_{deploy}) i.e., $T_{recovery} = T_{detect} + T_{select} + T_{deploy}$.

Further, we also prove that OCUS achieves Pareto optimality when coalition-switching energy costs (z_{ik}^{switch}) are integrated into the payoff function (Eq. (31)).

Let the payoff of UAV u_i in coalition λ_j be given by the extended form of Eq. (31), i.e., $\pi_i(\lambda_j) = \alpha_1 \times U(u_i, t_k) + \alpha_2 \times R_S(uo_i) + \alpha_3 \times T_A(uo_i) - \alpha_4 \times C_R(uo_i) - \alpha_5 \times z_{ik} - \alpha_6 \times z_{ik}^{switch}$, where $z_{ik}^{switch} \geq 0$ represents the energy cost incurred if u_i switches into coalition λ_j . To accurately model real-world energy penalties and prevent oscillatory behavior, we incorporate a memory-aware switching cost defined as:

$$z_{ik}^{switch} = \begin{cases} z_{ik}, & \text{if coalition } \lambda_j \text{ is new for } u_i, \\ \eta \times z_{ik}, & \text{if } \lambda_j \text{ was visited by } u_i \text{ within a recent time window,} \end{cases} \quad (38)$$

where $0 \leq \eta < 1$ is a discount factor reflecting reduced energy expenditure upon rejoining recently left coalitions.

As Eq. (33) ensures $\pi_i(\lambda_j)$ is strictly concave in the coalition index λ_j , i.e., $\partial^2 \pi_i / \partial \lambda_j^2 < 0$, this memory-aware extension preserves strict concavity, as the switching cost term enters linearly and its second derivative is zero. Thus, any NE of the OCUS algorithm yields a coalition assignment that is Pareto optimal, i.e., no UAV can improve its payoff without decreasing the payoff of another UAV.

Table 2
Theoretical comparison of OCUS with existing approaches.

Feature	OCUS (Proposed)	MSCGTO [27]	FedBarg [28]	Decentralized [37]
Multi-owner UAV coordination	✓	×	×	×
Adaptivity to UAV Heterogeneity	✓	×	×	✓
Efficient coalition time & size	✓	✓	✓	✓
Sensor-task matching	✓	×	×	×
Energy-aware coalition formation	✓	×	✓	×
Reputation-based coalition	✓	×	×	×
Task allocation fairness	✓	×	×	×
Cost competitiveness consideration	✓	×	✓	×
Payoff function with NE stability	✓	✓	✓	×

Table 3
Simulation parameters.

Parameter	Values
Simulation area	50×50 unit ²
Simulation distance	1–50 unit
Number of tasks	1–15
Number of UAVs	1–25
Number of end-users	1–5
Maximum price charged per UAV by owner	200 unit
Price charged per UAV by owner	10–200 unit
Energy consumed per task by UAV	10–250 units (in Joules)
Task required utility	10–30 units
UAV contribution value	10–30 units

By definition of a Nash equilibrium, no UAV can unilaterally switch coalitions to increase its payoff. In particular, for each UAV u_i currently in coalition λ_j , we have $\pi_i(\lambda_j) \geq \pi_i(\hat{\lambda})$ for every other coalition $\hat{\lambda}$, where π_i now includes the memory-aware switching-cost term. The first-order optimality condition (Eq. (32)) holds with switching cost, and the second-order condition (Eq. (33)) remains valid because $\frac{\partial^2 z_{\text{switch}}}{\partial \lambda_i^2} = 0$. Hence, strict concavity is preserved.

Suppose, for contradiction, that the equilibrium outcome were not Pareto optimal. Then, there would exist an alternate coalition $\hat{\lambda}$ for some UAV u_i that strictly increases π_i without decreasing any other UAV's payoff. Formally, assume there exists $\hat{\lambda} \neq \lambda_j$ such that $\pi_i(\hat{\lambda}) > \pi_i(\lambda_j)$ and $\pi_k(\hat{\lambda}_k) \geq \pi_k(\lambda_k)$ for all $k \neq i$, where $\hat{\lambda}_k$ denotes the coalition of UAV u_k under the hypothetical reallocation. This contradicts the equilibrium condition because any unilateral deviation must necessarily cause a loss to at least one other UAV due to the strict concavity preserved by the memory-aware switching cost.

Algorithm 2 Evaluate Best Coalition

Input: UAV u_i , Set of coalitions Λ

Output: $(\lambda_{\text{best}}, \pi_{\text{best}})$

```

1: Initialize  $\pi_{\text{best}} \leftarrow -\infty$ ,  $\lambda_{\text{best}} \leftarrow \text{null}$ 
2: for each coalition  $\lambda_j \in \Lambda$  do
3:   Temporarily assign  $u_i$  to  $\lambda_j$ 
4:   Compute payoff  $\pi_i(\lambda_j)$  using Eq. (31)
5:   if  $\pi_i(\lambda_j) > \pi_{\text{best}}$  then
6:      $\pi_{\text{best}} \leftarrow \pi_i(\lambda_j)$ 
7:      $\lambda_{\text{best}} \leftarrow \lambda_j$ 
8:   end if
9: end for
10: return  $(\lambda_{\text{best}}, \pi_{\text{best}})$ 

```

5. Performance evaluation

5.1. Simulation design

This section presents the performance analysis of the proposed scheme, OCUS. We conduct simulation using Algorithms 1 and 2. We first describe the simulation design, followed by a detailed analysis

and discussion of the results. The simulation experiments were conducted across 100 iterations and involves 1 to 25 heterogeneous UAVs. In our simulation, UAVs are randomly distributed across a 50×50 unit² area. Their (x, y, z) coordinates are generated using a uniform distribution, $x \sim \mathcal{U}(0, 50)$, $y \sim \mathcal{U}(0, 50)$, $z \sim \mathcal{U}(5, 20)$. Each UAV is randomly assigned between 5 to 15 sensors, chosen from a predefined set. Initial energy levels are sampled from a range of [50, 200] Wh, and average power consumption is drawn from [100, 120] W. Clock speed and memory are selected from ranges [5, 26] MHz and [2, 17] GB, respectively.

Tasks are randomly generated within the same spatial area, and each task specifies sensor requirements, maximum latency, and data volume, all sampled uniformly from realistic bounds. Task locations follow the same uniform distribution as UAVs, ensuring diverse deployment scenarios. The details of simulation parameter is shown in Table 1. During the simulation, UAVs follow the Random Waypoint mobility model [38]. Table 3 lists the simulation parameters.

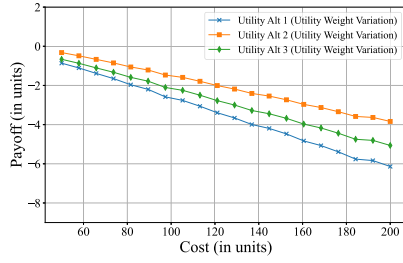
Benchmark Schemes: (i) Decentralized coalition formation of multiple UAVs in an uncertain region (Decentralized) [37]: This work presents an approach to form coalitions of heterogeneous UAVs for collaborative search-and-attack missions against stationary ground targets. The method focuses on forming the smallest possible coalition to meet the resource requirements and neutralize a target in the least amount of time. UAVs are sorted based on resource efficiency, and coalitions are selected by considering resource adequacy and arrival times. (ii) Multi-Service Edge Computing Management With Multi-Stage Coalition Game Task Offloading (MSCGTO) [27]: This work proposes the MSCGTO framework for efficient task offloading in dynamic multi-service edge computing environments. It enables service providers to form coalitions and offload computation to 5G-enabled edge servers, optimizing profit and user-perceived latency. The approach accounts for resource heterogeneity, user mobility, and service dynamics and includes distributed algorithms that balance system utility and delay. (iii) QoS-Aware Resource Bargaining for Federated Learning Over Edge Networks in Industrial IoT (FedBarg) [28]: This work introduces FedBarg, a game-theoretic approach for optimal resource allocation in federated learning over wireless edge networks in Industrial IoT settings. It addresses the challenges of constrained device resources, quality-of-service guarantees, and fair incentivization by modeling the problem as a Nash bargaining game. The proposed solution combines coalition formation and utility theory to enable partial contributions of computation and data from devices.

5.2. Analysis

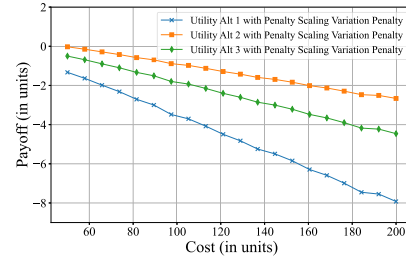
In this subsection, we analyze the computational complexity and communication overhead.

5.2.1. Computational complexity analysis

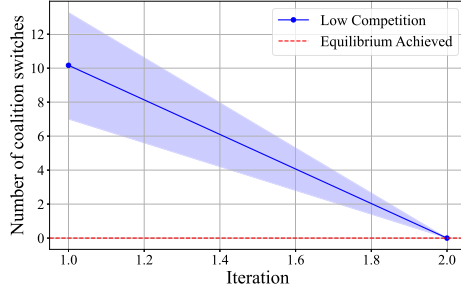
The proposed coalition formation algorithm exhibits a polynomial time complexity of $O(n^3)$. This complexity arises in the worst-case scenario, where each UAV must evaluate its payoff across all possible coalitions during each iteration. Specifically, Line 5 of Algorithm 1



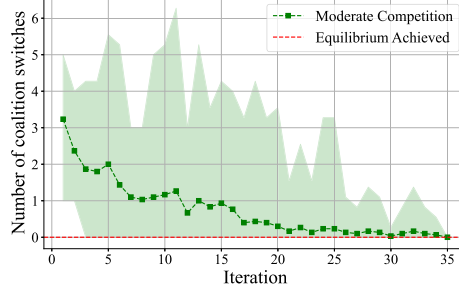
(a) Payoff vs Cost-Utility Weight Variation



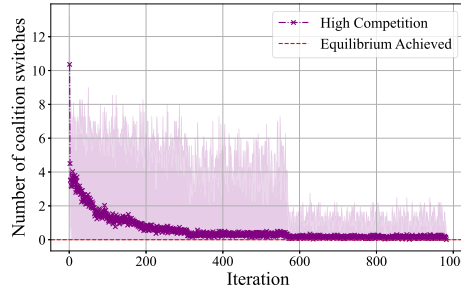
(b) Payoff vs Cost-Penalty Scaling Variation

Fig. 4. Comparison of various Payoff related parameters for different utility approaches.

(a) No. of coalition switches per iteration: Low Competition



(b) No. of coalition switches per iteration: Moderate Competition



(c) No. of coalition switches per iteration: High Competition

Fig. 5. Number of coalition switches required to reach equilibrium in various conditions.

performs n iterations over the UAVs. For each UAV, Line 7 invokes Algorithm 2, which iterates over all coalitions $\lambda_j \in \Lambda$, i.e., across n tasks, and computes the payoff for each (Line 2). Therefore, the inner subroutine has time complexity $O(n)$ per UAV. As a result, the overall execution includes $O(n)$ (outer loop over UAVs) $\times$ $O(n)$ (coalition evaluation inside subroutine) operations per iteration.

Assuming a maximum of I iterations to convergence (typically $I \leq n$ due to diminishing payoff property), the total time complexity becomes $O(I \times n^2)$. In the worst case, where $I = n$, this leads to an upper-bound time complexity of $O(n^3)$.

5.2.2. Communication overhead analysis

In the OCUS, coalition formation relies on distributed decision-making, where each UAV evaluates its payoff across multiple coalitions and communicates its intent to join or switch. This results in communication exchanges that we analyze here in terms of message complexity. n is the total number of UAVs and k is the number of tasks. During each iteration of coalition evaluation:

- Each UAV sends a coalition entry request to the coalition associated with each task ($O(n)$ messages per UAV).

- Coalition responses are returned based on payoff evaluation (confirmation or rejection) ($O(n)$ messages).
- If coalition switching occurs, the UAV broadcasts its updated coalition membership to others ($O(n)$ messages in the worst case).

Thus, each iteration can result in $O(n^2)$ messages in the worst case. Let I denote the number of convergence iterations (typically bounded and much smaller than n). The total communication overhead is therefore bounded by $O(I \times n^2)$.

5.3. Result and discussions

The result of several UAV strategies such as sensor match and proximity, balanced optimization, and cost competitiveness are shown in Fig. 2. We can see the sensor Match and proximity (OCUS-A) strategy works best because it reaches the maximum utility value among others strategies, followed by cost competitiveness (OCUS-C) and balanced optimization (OCUS-B). As we have seen in Algorithm 1, the payoff across strategies continues to increase until reaching an equilibrium,

where the maximum achievable payoff is obtained, as evident from Fig. 2. Experimentally, sensor match and proximity (OCUS-A) strategy dominates over the other two in the payoff calculation, as without having required set of sensors the utility of the coalition of UAVs drastically decreases.

Fig. 3(a) shows the task completion rate with increase in distance between UAV and location of task for UAV coalition of varying number of UAVs. The result shows that, for a given number of UAVs, the task completion rate is initially 100% up to a threshold value for distance. Here, threshold refers to the maximum distance up to which the task completion rate remains 100% – that is, UAVs successfully complete all tasks without any degradation in performance. However, the task completion rate begins to decline after the threshold, as it becomes difficult for UAVs to perform their task as distance increases. Furthermore, it is evident that the threshold value increases marginally for larger coalitions of UAVs but decreases for smaller coalitions. Fig. 3(b) represents the variation in utility with increase in distance between UAV and location of task for UAV coalition of varying number of UAVs. We observe, the utility for UAV coalitions initially decreases drastically. This happens because as seen in the Eq. (19), the distance is inversely related to the utility factor and we obtain a plot similar to a hyperbola shape. Fig. 3(c) shows the relation between distance and energy expenditure per UAV. As distance increases, more energy is consumed traveling to the application location and hence energy uses increases. With the decreasing number of UAVs, the number of task per UAV will increase. Consequently, a UAV's energy consumption increases as it services more tasks.

The comparison of payoff and energy cost with different variations from components of the Eq. (31) is shown in Fig. 4. Fig. 4(a) shows three alternative (Alt.) variations of the weight distribution — Alt. 1 signifies heavy penalties as the coefficient of energy penalty is 2.5 times that of Alt. 2, Alt. 2 represents the case of low penalties and hence has the least impact from rising cost, and Alt. 3 signifies the condition where penalty and utility are balanced, hence having coefficient here as 1.5 times that of Alt. 2. We observe that payoffs are highly sensitive to penalty scaling, higher penalty amplify the impact of rising costs as we can see in plot for Alt. 1, while Alt. 2 rise is highest as it has the lowest penalty. This also signifies that our algorithm is able to cope up with increasing costs as descent in payoff is not very steep.

Fig. 4(b) shows three alternative variations of the penalty distribution — Alt 1 with 80% weight to utility and 10% to reputation and task allocation fairness each, Alt 2 with 80% weight to reputation and 10% to utility and fairness each, Alt 3 with 60% weight to utility and 20% to reputation and fairness each. We observe that payoffs are higher when utility is not the dominating factor, as utility decreases steeply when cost increases, hence reputation biased approaches are much likely to attain higher payoff.

Fig. 5 shows the number of coalition switches required to achieve equilibrium with number of iterations and with changing competition which is based on number of UAVs and coalition switching environment. We have iterated over 30 random seeds and taken the average from them. Fig. 5(a) shows OCUS reaches the Nash Equilibrium and gets the max payoff in just 2 iterations with minimal coalition switching. Fig. 5(b) shows the effect of increased competition among UAVs on coalition formation. When there are more UAVs and the payoff values are very similar, the algorithm requires more iterations to reach equilibrium. Although the process slows down slightly due to higher competition, the algorithm efficiently manages this complexity, ensuring it reaches equilibrium. Despite this, OCUS ensures that the number of coalition switches per iteration remains relatively low, typically within the range of 2 to 4. This highlights the algorithm's efficiency in managing coalition dynamics while converging to an optimal solution under competitive conditions. Fig. 5(c) shows the worst-case scenario when the algorithm has to check for almost every possible coalition and hence take multiple coalition switches over a large number of iterations.

Across a range of network sizes, the convergence behavior of OCUS, as illustrated in Fig. 6, shows predictable coalition formation and repair dynamics. The algorithm's $\mathcal{O}(n^3)$ complexity scaling is confirmed by the initial formation's exponential decay patterns, which converge at about 820 ms for 25 UAVs, 1100 ms for 50 UAVs, and 1600 ms for 100 UAVs. All repair operations correctly begin from the stable coalition baseline (~ 300 ms) in Fig. 6, which measures coalition repair latency under various failure scenarios. Because localized adjustments maintain the overall coalition structure, a single UAV failure in a 50-UAV network converges in less than 20 iterations. Although it takes about 30 iterations, the 5% failure scenario (5 UAVs in a 100-UAV network) recovers much more quickly than full reformation.

The 20% correlated failure scenario provides crucial insight, exposing a two-phase behavioral pattern that confirms the Real-Time Coalition Repair Protocol's theoretical bounds. The system attempts standard Nash Equilibrium repair from iterations 0–60, cutting the time from 2000 ms to 900 ms. However, a sudden discontinuity that jumps to 1200 ms at iteration 60 – the crucial decision point – indicates the activation of the reformation penalty. This behavior establishes clear operational boundaries for real-world UaaS deployments by showing that, above a failure threshold of 15%–20%, marginal payoff conditions (Eqs. (32)–(33)) fail, Pareto optimality through incremental repairs is compromised, and full reformation becomes more efficient than continued repair attempts.

To validate the robustness of our reputation system, we simulated adversarial scenarios in which a varying percentage of UAV owners act maliciously. These actors attempt to compromise the system by submitting dishonest feedback, engaging in data poisoning, and launching Sybil attacks. We compared the performance of the original OCUS scheme with an enhanced version that integrates a Byzantine Fault Tolerant (BFT) consensus mechanism [39] through a Trusted Reputation Aggregator (TRA). This extended OCUS with BFT framework incorporates a quorum-based validation protocol before any reputation update is accepted. Specifically, for each task completion report, a randomly selected committee of N validators – drawn from a pool of high-reputation UAV owners – must agree on the reported metrics. The report is considered valid only if at least $f + 1$ validators reach consensus, where $f = \lfloor (N - 1)/3 \rfloor$ is the maximum number of malicious participants the system can tolerate. Reports that fail to achieve consensus are discarded, and senders of conflicting or unverifiable reports are penalized via a reputation deduction. Fig. 8 illustrates the results of this robustness experiment, with each point representing an average across 100 independent runs at different malicious actor percentages. The payoff under the OCUS without BFT degrades rapidly as the number of dishonest actors increases, indicating vulnerability to manipulated data. In contrast, the OCUS with BFT maintains a consistently high and stable payoff, demonstrating that the consensus-enforced TRA effectively filters out falsified inputs and upholds system integrity. These findings affirm that the BFT-enhanced approach ensures resilient coalition formation, even in the presence of adversarial behaviors. On the other hand, BFT mechanism introduces measurable communication overhead that increases quadratically with the number of validators, despite its strong defense against malevolent actors. To tackle this, we propose threshold signature schemes for lightweight cryptographic attestation in order to overcome this limitation. Specifically, we implement a k -out-of- m Elliptic Curve Digital Signature Algorithm (ECDSA) where reputation updates require attestations from at least $k = \lceil \frac{2m}{3} \rceil + 1$ validators out of m available high-reputation UAVs.

We conducted a Monte Carlo sensitivity analysis, as shown in Fig. 7, to assess the robustness of the utility function's scaling constants ($\alpha_1, \alpha_2, \alpha_3$) in Eq. (19). Over 10,000 iterations, each parameter was sampled from a normal distribution within its domain. The results indicate that the utility function is robust and responds predictably to parameter variations. Correlation coefficients (r) ranged from 0.153 to 0.762, indicating moderate to strong linear relationships between each parameter and the resulting utility score. Specifically, the sensor match

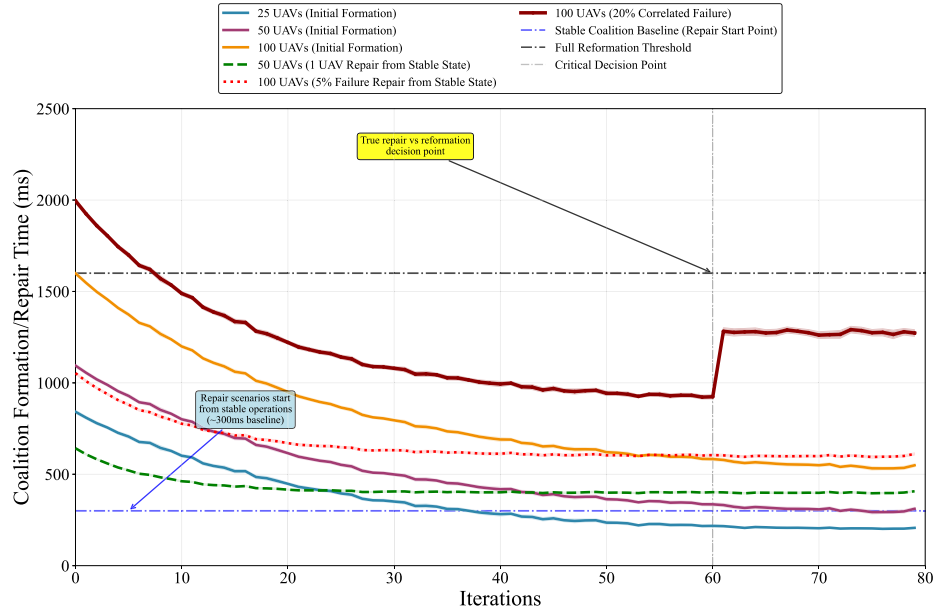


Fig. 6. Coalition formation time.



Fig. 7. Monte carlo analysis for utility function.

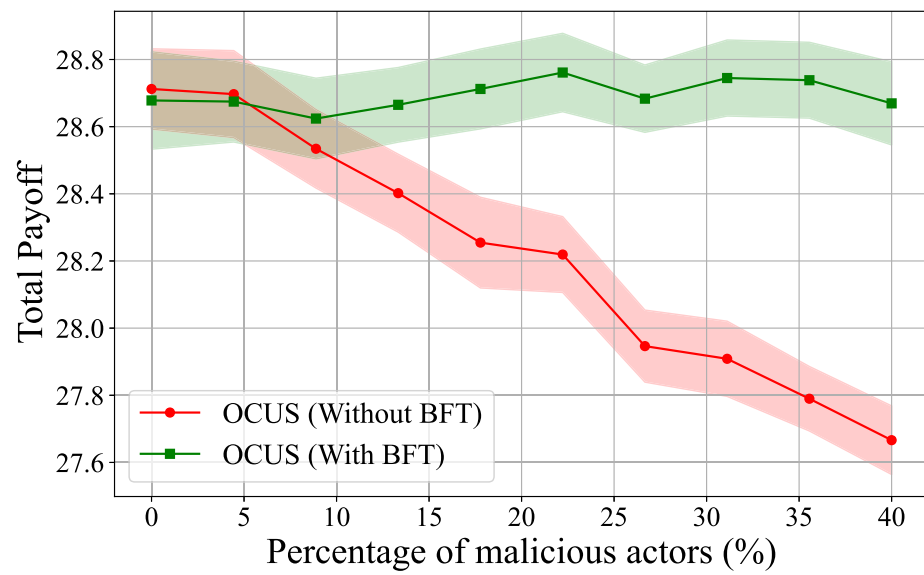


Fig. 8. Robustness of OCUS with and without BFT.

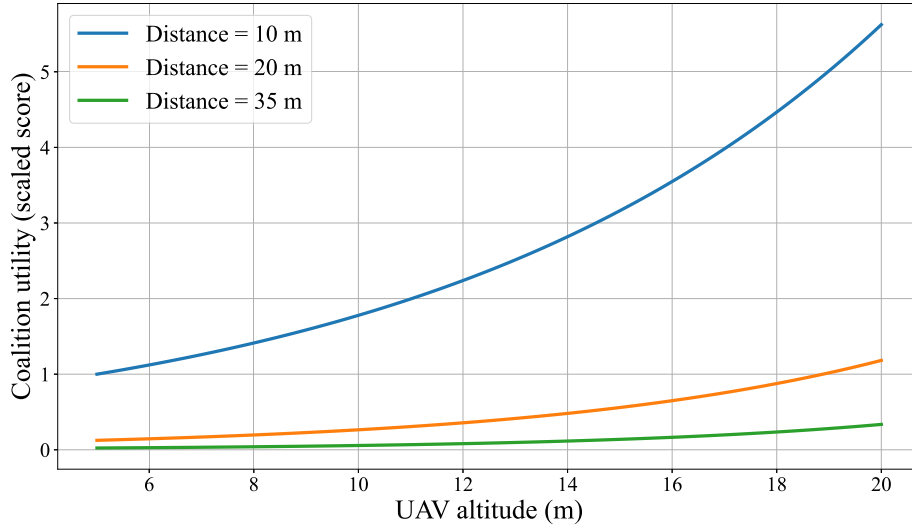
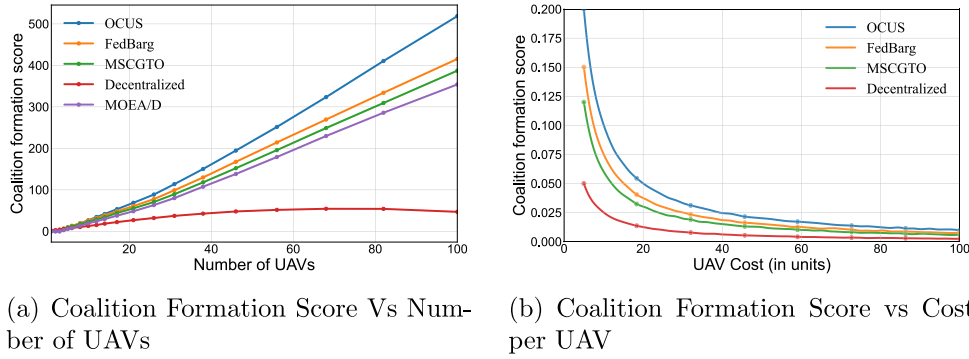


Fig. 9. Effect of UAV altitude on coalition utility.



(a) Coalition Formation Score Vs Number of UAVs

(b) Coalition Formation Score vs Cost per UAV

Fig. 10. Comparison of Coalition Formation Score of OCUS with other benchmark schemes.

weight (α_1) showed the strongest influence ($r = 0.762$), followed by task execution agility (α_2 , $r = 0.378$), and proximity (α_3 , $r = 0.153$). Further, utility values remained stable within an acceptable range (0.2 to 0.9) across all simulations, confirming that minor changes in parameter values do not lead to unpredictable or unstable outcomes. The scatter plots further demonstrate predictable, linear trends, reinforcing the mathematical consistency of the model. Overall, the sensitivity ranking – sensor match > agility > proximity – aligns with domain expectations, where accurate sensor-task alignment is typically the most critical factor in UAV selection. This analysis confirms that the utility function is robust and suitable for real-world UaaS deployments, particularly in scenarios where exact tuning of weight parameters may not be feasible.

Fig. 9 illustrates how coalition utility varies in dense urban environments, where 3D positioning and signal attenuation strongly influence a UAV's value to its coalition. Using the extended Effective Proximity Factor (EPF) with the Okumura–Hata model, two clear patterns emerge. First, across all tested horizontal distances (10 m, 20 m, and 35 m), utility increases sharply as UAV altitude rises from 5 m to 20 m within the urban canyon. This trend reflects the reduced path loss once the UAV clears street-level obstacles, improving link quality. Second, proximity proves to be a dominant factor: a UAV only 10 m away achieves much higher utility than one at 35 m, as both distance-related signal attenuation and the inverse proximity relationship penalize farther UAVs. These results confirm that the enhanced OCUS model realistically captures the combined effects of altitude and distance in urban deployments, favoring UAVs that are both higher and closer to tasks to form more robust and reliable coalitions.

We compare OCUS with existing state-of-the-art approaches. Fig. 10(a) compares OCUS with Decentralized algorithm [37], FedBarg [28], MSCGTO [27], and MOEA/D [40] for forming coalitions with number of UAVs which range from 1–100, showcasing scalability. The fixed cost per UAV and a fixed resource value by UAV are 4 units and 10 units. We observe that OCUS calculates coalition utility in each step and avoids including UAVs, which do not form efficient coalitions with respect to their payoff value. A coalition is not formed, and the score remains zero until the minimum required number of UAVs, defined as the threshold, is reached for forming an optimal coalition. Once the threshold is reached, OCUS begins forming efficient coalitions and drastically increases its score, demonstrating superior scalability compared to all baseline approaches. This threshold strategy ensures that OCUS avoids forming inefficient coalitions that would waste resources and reduce overall system performance. Whereas FedBarg shows competitive performance, demonstrating the effectiveness of its bargaining-based approach, while MSCGTO performs moderately well with its multi-stage coordination mechanism. The Decentralized approach shows the least scalability, exhibiting a concerning trend where performance begins to plateau and eventually declines at larger fleet sizes. The primary reason for this performance decline lies in the lack of global optimization perspective in decentralized systems. As the number of UAVs increases, the decentralized approach suffers from coordination chaos where each UAV makes locally optimal decisions without considering the broader system implications. This leads to redundant resource allocation, where multiple UAVs compete for the same tasks or geographical areas, resulting in inefficient utilization of the expanded fleet capacity. On the other hand, MOEA/D, a leading

decomposition-based multi-objective algorithm, takes a generic but powerful approach to coalition formation. MOEA/D handles moderate scaling well – forming coalitions as UAV numbers rise – yet its performance consistently lags behind OCUS. This is mainly because MOEA/D lacks UAV-specific modeling, such as task needs, preferences, or coalition stability, which limits its effectiveness in complex, heterogeneous settings. As fleet size increases, its performance plateaus or even drops due to difficulty maintaining solution diversity. Unlike OCUS, it cannot exploit domain-specific strategies like payoff balancing or stability via Nash equilibrium. While MOEA/D remains a strong general-purpose optimizer, it proves less efficient and stable for large-scale UAV coalition formation. Compared to MOEA/D, OCUS offers a favorable trade-off between computational speed and convergence reliability. While MOEA/D explores a broader Pareto front, it incurs higher overhead as the UAV fleet scales. OCUS, with its tractable worst-case complexity of $\mathcal{O}(n^3)$ and task-aware, equilibrium-driven design, achieves faster convergence and stable coalition formation. This makes it suitable for real-time large-scale UaaS deployment, despite modest trade-offs in solution diversity.

Fig. 10(b) presents the cost analysis, illustrating that OCUS consistently outperforms all baseline methods across the entire range of costs. OCUS maintains the highest coalition formation scores due to its Nash Equilibrium-based optimization, which enables intelligent adaptation to cost constraints by forming smaller, more efficient coalitions that still satisfy task requirements. FedBarg's performance degradation under cost constraints arises from its bargaining-based mechanism, which becomes increasingly inefficient as costs rise. The iterative bargaining process involves multiple rounds of negotiation among UAV owners, and as costs increase, the negotiations become more challenging, with UAVs showing reduced willingness to compromise. This bargaining overhead creates additional computational and communication costs that compound the original cost constraints, resulting in suboptimal coalition formations that prioritize individual UAV benefits over collective efficiency. For MSCGTO, as costs increase, the multi-stage process becomes prohibitively expensive because each stage involves separate optimization rounds with associated computational and communication overheads. Whereas the lack of optimization in the decentralized approach forces it to make suboptimal coalitions and results in redundant UAV usage and therefore a lesser coalition formation score. Table 2 presents a comparison between OCUS and existing schemes.

6. Conclusion

In this work, we proposed an optimal scheme, OCUS, for forming coalitions of UAVs from multiple UAV owners in the UaaS platform to complete a given application. Our approach addressed the challenges due to inefficient coalitions in UaaS. We defined a payoff function that accounts for various factors, such as the UAV's utility, the UAV owner's reputation score, the competitiveness ratio, the fairness factor of task allocation, and the energy consumption for the task. Using Nash Equilibrium, OCUS ensure stable coalition formations where UAVs maximize their payoffs without deviating. We also covered a subproblem of selecting the best UAV owners from factors like Reputation score, cost competitiveness ratio, and task allocation factor. Extensive simulations demonstrated the superiority of OCUS compared to the benchmark approaches, showing significant improvements in coalition stability and coalition formation score. In the future, this work can be expanded to expand the scope of the OCUS to include inter-platform collaboration, enabling cross-platform UAV coalitions for large-scale operations. It can also incorporate redundancy-aware strategies, such as backup UAV provisioning to handle mid-application failures.

CRediT authorship contribution statement

Prince Kumar: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Akhand Pratap Narayan Singh:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anchal Dubey:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Farid Nait-Abdesselam:** Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis. **Arijit Roy:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] A. Heidari, N. Jafari Navimipour, M. Unal, G. Zhang, Machine learning applications in internet-of-drones: Systematic review, recent deployments, and open issues, *ACM Comput. Surv.* 55 (12) (2023) 1–45.
- [2] A. Ahmed, C. Chaieb, W. Ajib, H. Elbiaze, R. Glitho, UAVs deployment optimization in cell-free aerial communication networks, *Comput. Commun.* 232 (2025) 108041, <http://dx.doi.org/10.1016/j.comcom.2024.108041>, URL <https://www.sciencedirect.com/science/article/pii/S0140366424003888>.
- [3] A.F. Mostafa, M. Abdel-Kader, Y. Gadallah, A UAV-based coverage gap detection and resolution in cellular networks: A machine-learning approach, *Comput. Commun.* 215 (2024) 41–50, <http://dx.doi.org/10.1016/j.comcom.2023.12.010>, URL <https://www.sciencedirect.com/science/article/pii/S0140366423004450>.
- [4] Z. Shah, U. Javed, M. Naeem, S. Zeadally, W. Ejaz, Mobile edge computing (MEC)-enabled UAV placement and computation efficiency maximization in disaster scenario, *IEEE Trans. Veh. Technol.* 72 (10) (2023) 13406–13416, <http://dx.doi.org/10.1109/TVT.2023.3274107>.
- [5] S. Kumari, S. Srirangarajan, Multi-UAV path planning for connectivity-based sweep coverage, *Ad Hoc Netw.* 178 (2025) 103966, <http://dx.doi.org/10.1016/j.adhoc.2025.103966>, URL <https://www.sciencedirect.com/science/article/pii/S1570870525002148>.
- [6] D. Dissanayaka, T.R. Wanasinghe, O. De Silva, A. Jayasiri, G.K.I. Mann, Review of navigation methods for UAV-based parcel delivery, *IEEE Trans. Autom. Sci. Eng.* 21 (1) (2024) 1068–1082, <http://dx.doi.org/10.1109/TASE.2022.3232025>.
- [7] A. Andreadis, G. Giambene, R. Zambon, Design and analysis of a lora-based system with scheduled transmissions from IoT nodes to UAV in rural areas, *Ad Hoc Netw.* 175 (2025) 103868, <http://dx.doi.org/10.1016/j.adhoc.2025.103868>, URL <https://www.sciencedirect.com/science/article/pii/S1570870525001167>.
- [8] J. Yapp, R. Seker, R. Babiceanu, UAV as a eervice: Enabling on-demand access and on-he-fly re-tasking of multi-tenant UAVs using cloud services, in: The 35th IEEE/AIAA Digital Avionics Systems Conference, DASC, 2016, pp. 1–8, <http://dx.doi.org/10.1109/DASC.2016.7778007>.
- [9] F. Nait-Abdesselam, A. Alsharoa, M.Y. Selim, D. Qiao, A.E. Kamal, Towards enabling unmanned aerial vehicles as a service for heterogeneous applications, *J. Commun. Netw.* 23 (3) (2021) 212–221, <http://dx.doi.org/10.23919/JCN.2021.000015>.
- [10] N. Pathak, S. Misra, A. Mukherjee, A. Roy, A. Zomaya, UAV virtualization for enabling heterogeneous and persistent UAV-as-a-service, *IEEE Trans. Veh. Technol.* 69 (6) (2020) 6731–6738, <http://dx.doi.org/10.1109/TVT.2020.2985913>.
- [11] A. Roy, V.M.R. Tummala, V. Yadam, Serv-HU: Service hand-off for UAV-as-a-service, *IEEE Trans. Serv. Comput.* 18 (1) (2025) 414–426, <http://dx.doi.org/10.1109/TSC.2024.3521684>.
- [12] P. Kumar, V.M.R. Tummala, A. Roy, TOFU: Trust-based task offloading in UAV-as-a-service, *IEEE Trans. Veh. Technol.* (2025) 1–12, <http://dx.doi.org/10.1109/TVT.2025.3604589>.
- [13] J.M. Staats, The cooperative as a coalition: a game-theoretic approach, *Am. J. Agric. Econ.* 65 (5) (1983) 1084–1089.

- [14] M. Yassin, H. Ould-Slimane, C. Talhi, H. Boucheneb, Multi-tenant intrusion detection framework as a service for saas, *IEEE Trans. Serv. Comput.* 15 (5) (2022) 2925–2938, <http://dx.doi.org/10.1109/TSC.2021.3077852>.
- [15] F.A. Al-Zahrani, Subscription-based data-sharing model using blockchain and data as a service, *IEEE Access* 8 (2020) 115966–115981, <http://dx.doi.org/10.1109/ACCESS.2020.3002823>.
- [16] A. Roy, S. Misra, S. Bandyopadhyay, A. Jamalipour, Mobile sensor-cloud for rendering sensors-as-a-service, *IEEE Syst. J.* 15 (4) (2021) 5174–5185, <http://dx.doi.org/10.1109/JSYST.2021.3091585>.
- [17] A. Roy, S. Misra, S. Nag, PRIME: An optimal pricing scheme for mobile sensors-as-a-service, *IEEE Trans. Mob. Comput.* 21 (4) (2022) 1362–1373, <http://dx.doi.org/10.1109/TMC.2020.3023885>.
- [18] C. Roy, S. Misra, J. Maiti, U. Chakravarty, Safe-serv: Energy-efficient decision delivery for provisioning safety-as-a-service, *IEEE Trans. Serv. Comput.* 15 (4) (2022) 1954–1966, <http://dx.doi.org/10.1109/TSC.2020.3026135>.
- [19] C. Roy, S. Misra, Safe-passé: Dynamic handoff scheme for provisioning safety-as-a-service in 5G-enabled intelligent transportation system, *IEEE Trans. Intell. Transp. Syst.* 22 (8) (2021) 5415–5425, <http://dx.doi.org/10.1109/TITS.2021.3054694>.
- [20] C. Roy, S. Misra, J.J.P.C. Rodrigues, U. Chakravarty, RegPrice: Region-based pricing scheme for provisioning safety-as-a-service in IoT applications, *IEEE Trans. Veh. Technol.* 70 (4) (2021) 3017–3026, <http://dx.doi.org/10.1109/TVT.2021.3063930>.
- [21] F. Hasić, J. De Smedt, S.v. Broucke, E. Serral, Decision as a service (daas): A service-oriented architecture approach for decisions in processes, *IEEE Trans. Serv. Comput.* 15 (2) (2022) 904–917, <http://dx.doi.org/10.1109/TSC.2020.2965516>.
- [22] J.M. Jorquera Valero, P.M. Sánchez Sánchez, M. Gil Pérez, A. Huertas Celdrán, G. Martínez Pérez, Trust-as-a-service: A reputation-enabled trust framework for 5g network resource provisioning, *Comput. Commun.* 211 (2023) 229–238, <http://dx.doi.org/10.1016/j.comcom.2023.09.010>, URL <https://www.sciencedirect.com/science/article/pii/S0140366423003201>.
- [23] F. Maleki, M.A.J. Jamali, A. Heidari, Unmanned aerial vehicle routing based on frog-leaping optimization algorithm, *Sci. Rep.* 15 (1) (2025) 11249.
- [24] A. Heidari, N.J. Navimipour, M. Unal, A secure intrusion detection platform using blockchain and radial basis function neural networks for internet of drones, *IEEE Internet Things J.* 10 (10) (2023) 8445–8454.
- [25] S. Hadjres, N. Kara, M.E. Barachi, F. Belqasmi, An SLA-aware cloud coalition formation approach for virtualized networks, *IEEE Trans. Cloud Comput.* 9 (2) (2021) 475–491, <http://dx.doi.org/10.1109/TCC.2018.2865737>.
- [26] R. Kakkar, R. Gupta, S. Tanwar, J.J.P.C. Rodrigues, Coalition game and blockchain-based optimal data pricing scheme for ride sharing beyond 5G, *IEEE Syst. J.* 16 (4) (2022) 6321–6327, <http://dx.doi.org/10.1109/JSYST.2021.3126620>.
- [27] C.-C. Lin, Y. Chiang, H.-Y. Wei, Multi-service edge computing management with multi-stage coalition game task offloading, *IEEE Trans. Netw. Serv. Manag.* 21 (3) (2024) 3278–3291, <http://dx.doi.org/10.1109/TNSM.2024.3358414>.
- [28] A. Chakraborty, S. Misra, QoS-aware resource bargaining for federated learning over edge networks in industrial IoT, *IEEE Trans. Netw. Sci. Eng.* 10 (5) (2023) 2769–2778, <http://dx.doi.org/10.1109/TNSE.2022.3206474>.
- [29] T. Pinto, M. Wooldridge, Z. Vale, Consumer flexibility aggregation using partition function games with non-transferable utility, *IEEE Access* 9 (2021) 51519–51535, <http://dx.doi.org/10.1109/ACCESS.2021.3069416>.
- [30] F. Zafari, K.K. Leung, D. Towsley, P. Basu, A. Swami, J. Li, Let's share: A game-theoretic framework for resource sharing in mobile edge clouds, *IEEE Trans. Netw. Serv. Manag.* 18 (2) (2021) 2107–2122, <http://dx.doi.org/10.1109/TNSM.2020.3044870>.
- [31] R. Gupta, J. Gupta, Future generation communications with game strategies: A comprehensive survey, *Comput. Commun.* 192 (2022) 1–32, <http://dx.doi.org/10.1016/j.comcom.2022.05.024>, URL <https://www.sciencedirect.com/science/article/pii/S0140366422001761>.
- [32] Y. Wang, L. Zhang, C. Wei, Y. Tang, Joint optimization of resource allocation and computation offloading based on game coalition in C-V2X, *Ad Hoc Netw.* 150 (2023) 103266, <http://dx.doi.org/10.1016/j.adhoc.2023.103266>, URL <https://www.sciencedirect.com/science/article/pii/S1570870523001865>.
- [33] M. Asadi, M.A.J. Jamali, A. Heidari, N.J. Navimipour, Botnets unveiled: A comprehensive survey on evolving threats and defense strategies, *Trans. Emerg. Telecommun. Technol.* 35 (11) (2024) e5056.
- [34] A. Ribeiro, J.B. da Costa, G.P.R. Filho, L.A. Villas, D.L. Guidoni, S. Sampaio, R.I. Meneguette, HARMONIC: Shapley values in market games for resource allocation in vehicular clouds, *Ad Hoc Netw.* 149 (2023) 103224, <http://dx.doi.org/10.1016/j.adhoc.2023.103224>, URL <https://www.sciencedirect.com/science/article/pii/S1570870523001440>.
- [35] W. Peukert, Über die Abhängigkeit der Kapazität von der Entladestromstärke bei Bleiakкумуляtoren, *Elektrotech. Z.* 18 (1897) 287–288.
- [36] M. Hata, Empirical formula for propagation loss in land mobile radio services, *IEEE Trans. Veh. Technol.* 29 (3) (2013) 317–325.
- [37] S.A. Ali, X. Gao, X. Fu, Decentralized coalition formation of multiple UAVs in an uncertain region, in: *IEEE Advanced Information Management, Communicates, Electronic and Automation Control Conference, IMCEC, 2016*, pp. 916–920, <http://dx.doi.org/10.1109/IMCEC.2016.7867344>.
- [38] D.B. Johnson, D.A. Maltz, *Dynamic source routing in ad hoc wireless networks*, in: *Mobile Computing*, Springer, 1996, pp. 153–181.
- [39] W. Zhong, C. Yang, W. Liang, J. Cai, L. Chen, J. Liao, N. Xiong, Byzantine fault-tolerant consensus algorithms: A survey, *Electronics* 12 (18) (2023) 3801.
- [40] Q. Zhang, H. Li, MOEA/D: A multiobjective evolutionary algorithm based on decomposition, *IEEE Trans. Evol. Comput.* 11 (6) (2007) 712–731.